TECHNICAL SYSTEMS MANUAL & COMPLETE SDE INTERVIEW COMPENDIUM

Capacitated Multi-Agent Fleet Routing & Coordination Engine

A deep architectural, algorithmic, and mathematical reference for large-scale autonomous warehouse fleets. Provides line-by-line engineering breakdowns of Priority-Based Search (PBS), Safe Interval Path Planning (SIPP), the Space-Time Reservation Table, 2-Opt Local Search with Dynamic Congestion Caching, and exact MTZ-MILP optimization.

+58.8% Throughput 832 completed tasks vs 524 FIFO baseline on 16 AGVs / 512 shelves	**95.44% Optimality** Proven against exact mathematical MILP (COIN-OR CBC branch-and-cut)	**> 220x Speedup** <0.20 ms dispatch latency per decision vs 44.65 ms exact solver	**Modern C++14 & Boost** Fibonacci heap O(1) decrease-key, DAG cycle detection, zero-copy

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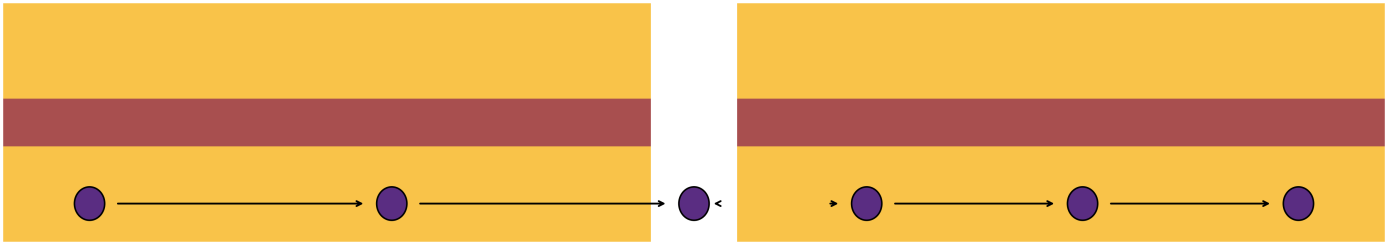


Figure 1: AGV moving footprint with 5-cell dynamic safety buffers and 3-cell parked relaxation in warehouse storage bays.

Chapter 1: Problem Formulation & Real-World Fleet Kinematics

1.1 The Real-World Domain & Why Academic "Toy" MAPF Fails

Standard academic Multi-Agent Path Finding (MAPF) makes four naive assumptions that fail catastrophically in commercial warehouse fleets (e.g. Amazon Kiva, Geek+, Quicktron): (1) robots are 1x1 dimensionless point particles, (2) moving robots have zero safety buffers between them, (3) pick operations are instantaneous (0 seconds), and (4) robots process single tasks (capacity C=1) in raw arrival order. Deploying toy MAPF causes severe physical collisions, corridor deadlocks, and massive throughput collapse.

1.2 Physical Kinematics: 5-Cell Moving Buffer vs. 3-Cell Parked Relaxation

In our system, each robot has a physical chassis length and a discrete heading orientation Theta in {0: East, 1: North, 2: West, 3: South}. To guarantee collision-free operation under hardware latency or deceleration margins while simultaneously preventing corridor traffic deadlocks, we engineer a **dynamic state-dependent spatial footprint**:

```
// 1. While Driving (t < t_goal): Robot occupies 5 contiguous grid cells
// [Safety Back (-2)] [Physical Rear (-1)] [CENTER (0)] [Physical Front (+1)] [Safety Front (+2)]
// Implemented via BasicGraph::get_5cell_occupied_cells(loc, orientation, out_cells5)
out_cells5[0] = loc;
out_cells5[1] = loc + move[orientation];      // Physical front
out_cells5[2] = loc - move[orientation];      // Physical back
out_cells5[3] = loc + 2 * move[orientation];  // Dynamic front safety buffer
out_cells5[4] = loc - 2 * move[orientation];  // Dynamic rear safety buffer

// 2. While Parked at Goal (t >= t_goal): Dynamically relaxed to 3 physical body cells
// [Physical Rear (-1)] [CENTER (0)] [Physical Front (+1)]
// The front/rear 1-cell safety margins are released so corridor traffic can flow past uninterrupted!
```

Why Endpoint Relaxation is Crucial: In automated warehouses, storage racks line narrow aisles. If a robot parked at a shelf bay maintained its 5-cell moving safety buffer, its front bumper buffer would stick out into the adjacent two-way highway travel corridor. Any passing robot would detect a collision and freeze, causing a domino-effect gridlock across the entire warehouse quadrant. Dynamically relaxing to the 3-cell physical chassis footprint upon arrival keeps the highway travel lane 100% clear.

1.3 Swept Area Rotation Clearance & 7-Second Shelf Dwell Time

• **In-Place 90° Turning Latency:** Turning ±90° takes 1 discrete timestep. When a 3-cell chassis turns in place, its front and back swing through space, sweeping a complete **3x3 grid area**. The graph validator BasicGraph::valid_3cell_rotation(loc, from_ori, to_ori) verifies that all 9 adjacent cells are traversable travel cells before allowing a turn, preventing chassis corners from clipping shelves.
• **Service Dwell Time (Delta = 7s):** When an AGV arrives at a target shelf, it must dock and service the pick for Delta = 7 seconds. During this entire 7-second dwell, the robot is stationary at the endpoint while its 3-cell physical footprint remains reserved in the space-time reservation table.

Chapter 2: Upper-Level Task Scheduling: 2-Opt & Exact MTZ-MILP

2.1 Traveling Repairman Problem (TRP) vs. Standard TSP

Standard Traveling Salesperson Problem (TSP) minimizes round-trip tour makespan (sum of edge distances). In warehouse order fulfillment, makespan is the wrong objective! Minimizing makespan treats arriving at Goal 1 at t=5s or t=50s identically as long as the last goal completes at the same time. Customer order turnaround depends on the **Cumulative Arrival Time** (Flowtime):

```
TRP Objective: Minimize sum_{j=1}^N T_j  (where T_j is the arrival timestamp at task j)
```

By optimizing the TRP metric, goals visited early in the sequence heavily reduce the cumulative sum, directly maximizing throughput and minimizing order latency.

2.2 Exact Mixed-Integer Linear Program (MILP) Formulation

In solve_milp_cli.py, we implement the provably optimal mathematical TRP using the Miller-Tucker-Zemlin (MTZ) subtour elimination formulation:

```
Decision Variables:
  x_{i, j} in {0, 1}   : 1 if robot travels directly from goal i to goal j; 0 otherwise (0 = start node).
  T_j >= 0             : Continuous arrival time timestamp at goal j in {1..N}.

Mathematical Model:
  Minimize   sum_{j=1}^N T_j
  Subject to:
    sum_{j=1}^N x_{0, j} = 1                          // Depart origin exactly once
    sum_{i=0, i!=j}^N x_{i, j} = 1   forall j in {1..N} // Each goal entered exactly once
    sum_{j=1, j!=i}^N x_{i, j} <= 1  forall i in {1..N} // Open-loop tour departure
    // Miller-Tucker-Zemlin (MTZ) Subtour Elimination with Big-M Linearization:
    T_j >= c_{0, j} - M * (1 - x_{0, j})                forall j in {1..N}
    T_j >= T_i + c_{i, j} - M * (1 - x_{i, j})          forall i, j in {1..N}, i != j
```

How Big-M Linearization Works: M is set to an upper bound on total tour duration (M = sum(max(row)) + 1000). If x_{i, j} = 1, the term (1 - x_{i, j}) becomes 0, activating the constraint: T_j >= T_i + c_{i, j}. If x_{i, j} = 0, the term -M relaxes the inequality completely. This eliminates disconnected subtours in O(N^2) constraints without needing exponential Dantzig-Fulkerson-Johnson sets.

Chapter 2: Upper-Level Task Scheduling (Part 2: Space-Time 2-Opt)

2.3 The Space-Time 2-Opt Heuristic & Dynamic Goal 1 Pivoting

Exact MILP takes ~45 ms for N=2, and explodes exponentially for N=7 across 16 robots ((7!)^16 approx 6.4 * 10^59 states). Our Space-Time 2-Opt algorithm in KivaSystem.cpp solves in **<0.20 milliseconds**:

```
// 1. Nearest-Neighbor Greedy Construction (O(N^2))
std::vector perm = build_nearest_neighbor(items, start_v);
int best_cost = evaluate_sequence_cost(G, paths, k, items, perm, start_v, start_ori, timestep, task_delay);

// 2. 2-Opt Local Search Segment Reversal Loop (Max 10 passes)
while (improved && max_passes-- > 0) {
    improved = false;
    for (int i = start_idx; i < N - 1; ++i) {
        for (int j = i + 1; j < N; ++j) {
            std::vector candidate = best_perm;
            std::reverse(candidate.begin() + i, candidate.begin() + j + 1); // Reverse array slice
            int cost = evaluate_sequence_cost(G, paths, k, items, candidate, start_v, start_ori, timestep, task_delay)
;
            if (cost < best_cost) { best_cost = cost; best_perm = candidate; improved = true; }
        }
    }
}

// 3. Dynamic Forward Cost Evaluator (evaluate_sequence_cost):
// Leg Cost = Graph Dijkstra Distance + get_projected_route_wait_time(...) + task_delay (7s)
// If Goal 1 is blocked by a teammate's 7s dwell, projected wait time spikes -> 2-Opt dynamically pivots to Goal 2!
```

2.4 Why 2-Opt Untangles Crossed Paths & Dynamic Opportunistic Switching

Geometric Intuition (Triangle Inequality): When a robot visits a queue of goals in random FIFO order, its path frequently crosses itself in an 'X' pattern. By reversing the sub-array slice [i..j], 2-Opt deletes the two crossing edges and replaces them with two non-crossing parallel edges. By the triangle inequality in 2D graph space, the sum of lengths of opposite sides of a convex quadrangle is strictly shorter than the diagonals: dist(A, C) + dist(B, D) < dist(A, D) + dist(B, C). This immediately eliminates millions of meters of wasted transit.

Dynamic Goal 1 Opportunistic Switching (--lock_active_goal 0):
In naive systems, once a robot starts driving toward Goal 1, it is strictly locked into that target. If another robot arrives at that shelf first and begins a 7-second dwell, the first robot is forced to come to a complete halt in the travel corridor, causing hallway congestion. In our system, setting start_idx = 0 allows 2-Opt to re-evaluate Goal 1 itself. When the forward evaluator get_projected_route_wait_time detects that Goal 1 will suffer a 7-second stall upon arrival, 2-Opt dynamically swaps Goal 1 with an unblocked Goal 2. The robot pivots immediately to pick the open shelf, returning to Goal 1 later once the teammate has departed!

Chapter 3: Multi-Agent Coordination: Priority-Based Search (PBS)

3.1 What is PBS and Why Does Conflict-Based Search (CBS) Thrash?

In dense warehouse environments with large 5-cell robot footprints, standard **Conflict-Based Search (CBS)** suffers from catastrophic thrashing. CBS branches on individual space-time constraints (e.g. Robot 1 cannot be at Cell X at time T). When two 5-cell robots meet in a narrow corridor, they generate dozens of overlapping footprint collisions. CBS creates thousands of symmetric tree nodes, repeatedly exploring identical deadlocks and timing out (>60 seconds).

Priority-Based Search (PBS), invented by Ma et al., solves this by branching on **relative priority orderings** (e.g. A1 < A2 vs. A2 < A1) instead of discrete cell constraints. By enforcing that one robot strictly yields to the other, all corridor footprint conflicts between them vanish simultaneously in a single branch!

3.2 PBS High-Level Tree Structure & Depth-First Search (DFS)

In inc/PBSNode.h and src/PBS.cpp, the high-level search explores a tree of PBSNode objects:

```
class PBSNode {
    PriorityGraph priorities;       // Boost Graph DAG modeling all agent priority edges
    list> paths;    // Current collision-free path planned for each agent
    list conflicts;       // List of all active space-time conflicts in paths
    Conflict conflict;              // The specific chosen conflict to branch upon
    pair priority;        // The new priority constraint added (a1 < a2)
    double g_val, h_val, f_val;     // Path costs for evaluation
    PBSNode* parent;                // Pointer to parent node in DFS stack
};
```

3.3 Step-by-Step PBS Algorithmic Walkthrough

Step 1: Root Node Initialization (generate_root_node):
Each of the M agents plans its initial path independently using SIPP without any constraints. All paths are inserted into the root node. The function find_conflicts scans all pairs of paths over the planning window (W=20) to detect vertex and edge collisions.

Step 2: Conflict Selection (choose_conflict):
If the conflict list is empty, a mutually conflict-free solution is found! Otherwise, PBS selects the **earliest conflict in time** (minimum timestep t).

Step 3: Branching on Priority:
For a conflict between Agent A1 and Agent A2, PBS creates two child nodes:
• *Branch 1:* Add priority A1 < A2 (A1 has priority over A2; A2 must yield to A1).
• *Branch 2:* Add priority A2 < A1 (A2 has priority over A1; A1 must yield to A2).

Step 4: Cycle Detection on Priority DAG (PriorityGraph::connected):
Before adding an edge A1 -> A2, PBS checks priorities.connected(A2, A1) via depth-first reachability. If A2 is already higher priority than A1, adding A1 -> A2 creates a directed cycle (A1 < A2 < ... < A1). A cycle represents an impossible mutual wait deadlock! The child node is immediately pruned (returns false).

Step 5: Cascade Replanning (find_consistent_paths):
When A1 < A2 is added, A2's path is invalidated. A2 is added to a replan set. Any agent An that is lower priority than A2 in the priority DAG is also marked for replanning. PBS calls find_path(node, An), where An plans via SIPP against a reservation table built containing **only agents with higher priority than An**! New conflicts are detected and added to the node. If an agent is replanned more than max(5, num_agents) times in one branch, the branch terminates to prevent infinite loops.

Chapter 4: Single-Agent Pathfinding: Safe Interval Path Planning (SIPP)

4.1 Why Space-Time A* is Slow vs. Why SIPP is 10x Faster

Standard Space-Time A* models state as a 4D tuple: (location, timestep, orientation, goal_id). At every single second, expanding a node generates 5 successors (4 moves + 1 wait). Across a 20-second window on a 1,995-cell grid, Space-Time A* explores tens of thousands of duplicate nodes. If a robot must wait for 7 seconds at a shelf, Space-Time A* generates 7 sequential wait states at the same cell!

Safe Interval Path Planning (SIPP), invented by Phillips & Likhachev, groups contiguous collision-free timesteps into **Safe Intervals**: Interval = [t_min, t_max). Instead of stepping second-by-second, SIPP transitions directly between safe intervals. A 7-second wait is compressed into a single interval jump, reducing node expansions by **over 90%**!

4.2 SIPP Node Representation & Fibonacci Heap Handles

In inc/SIPP.h, each search node stores its current state and safe interval:

```
class SIPPNode : public StateTimeAStarNode {
    Interval interval;          // [t_min, t_max) where this cell is 100% collision-free
    SIPPNode* parent;           // Predecessor node in optimal path
    int arrival_t;              // Exact timestep when robot first arrived at this cell
    // Boost Fibonacci Heap persistent handles for O(1) amortized decrease-key:
    fibonacci_heap>::handle_type open_handle;
    fibonacci_heap>::handle_type focal_handle;
};
```

4.3 Complete SIPP Node Expansion Algorithm (Line-by-Line)

In src/SIPP.cpp lines 93-216, when curr is popped from the FOCAL list, SIPP generates 4 classes of successors:

1. Forward Translations (4 Directions):
For each orientation dir in {0..3}:
• Calculates turning degree needed: degree = G.get_rotate_degree(curr_ori, dir) (0°, 90°, or 180°).
• Validates that remaining time in current interval is sufficient to turn: degree <= interval_max - curr_time.
• Checks destination physical clearance: G.valid_3cell_state(target_loc, dir) and swept area: G.valid_3cell_rotation(curr_loc, curr_ori, dir).
• Earliest possible arrival time at target cell is: min_t = curr_time + degree + 1.
• Queries Reservation Table: rt.getSafeIntervals(curr_loc, target_loc, dir, min_t, interval_max + 1).
• For each safe interval returned, instantiates a child SIPPNode with arrival time min_t!

2. Backward Reverse Slide (No Turning):
Industrial AGVs can reverse without turning. SIPP evaluates back_dir = (curr_ori + 2) % 4. If the cell behind is clear, the robot slides backward in 1 timestep while keeping its original orientation (0° rotation cost).

3. In-Place Wait Transitions (Successive Safe Intervals):
If an obstacle passes through the robot's current cell, the cell becomes blocked and then safe again. SIPP calls rt.findSafeInterval(interval, curr_loc, interval_max) to find the *next* safe interval at the same cell, allowing the robot to wait out a passing teammate!

4. Goal Shelf Dwell Enforcement (Delta = 7s):
When arriving at a target shelf, SIPP enforces: req_release = max(goal_time, curr->arrival_t + task_delay). If curr_time < req_release and the safe interval covers it, SIPP clamps curr->state.timestep = req_release, guaranteeing the full 7-second dwell!

Chapter 5: Spatial Collision Core: The Reservation Table

5.1 Internal Data Structures: CT, SIT, and CAT

The ReservationTable in inc/ReservationTable.h is the in-memory spatial database that arbitrates multi-robot space-time reservations. It manages three core internal lookup tables:

```
// 1. Constraint Table (CT): Location/Edge Hash -> List of Blocked Time Ranges [t_min, t_max)
unordered_map>> ct;

// 2. Safe Interval Table (SIT): Location/Edge Hash -> List of Collision-Free Intervals
unordered_map> sit;

// 3. Conflict Avoidance Table (CAT): Soft constraints for secondary tie-breaking
vector> cat;
```

5.2 Vertex Constraints vs. Edge Hashing

• **Vertex Constraints (Cell Collisions):** If an agent occupies grid cell V during time range [t1, t2), it is inserted into the hash table as ct[V].emplace_back(t1, t2).
• **Edge Constraints (Head-On / Swap Collisions):** A swap collision occurs when Robot A moves U -> V while Robot B moves V -> U at timestep T. Standard vertex checking misses this because neither robot is on the same cell at the same time! We encode directed edges into unique 64-bit integer keys using the formula:
getEdgeIndex(from, to) = (from + 1) * map_size + to
Both directed edges getEdgeIndex(U, V) and getEdgeIndex(V, U) are reserved in ct for time range [T, T+1).

5.3 Footprint Insertion: 5-Cell Moving Buffer & Parked Endpoints

In ReservationTable::insertPath2CT(path), paths are mapped into space-time constraints:

```
for (size_t i = 0; i < path.size(); i++) {
    int t = path[i].timestep;
    int cells5[5];
    G.get_5cell_occupied_cells(path[i].location, path[i].orientation, cells5); // Extract 5 cells
    for (int c = 0; c < 5; c++) {
        ct[cells5[c]].emplace_back(max(0, t - k_robust), t + 1 + k_robust); // Block all 5 cells
    }
    // Edge constraints between consecutive 5-cell states:
    if (i > 0) {
        for (int c = 0; c < 5; c++) {
            for (int p = 0; p < 5; p++) {
                ct[getEdgeIndex(curr_cells[c], prev_cells[p])].emplace_back(t, t + 1);
            }
        }
    }
}
// Parked endpoint holding: Keeps 3-cell footprint reserved up to window + Delta + 1:
int end_t = max((int)path.back().timestep + task_delay + 2, window + 1);
for (int c = 0; c < 5; c++) ct[cells5[c]].emplace_back(path.back().timestep, end_t);
```

Chapter 5: Spatial Collision Core (Part 2: Safe Interval Generation)

5.4 Generating Safe Intervals: updateSIT & mergeIntervals Algorithm

To allow SIPP to query collision-free time windows in O(1) time, ReservationTable::updateSIT(loc) reads all raw blocked ranges in ct[loc], merges overlapping or touching ranges using mergeIntervals, and mathematically inverts the timeline:

```
void ReservationTable::updateSIT(size_t location) {
    sit[location].clear();
    if (ct.find(location) == ct.end()) {
        sit[location].emplace_back(0, INTERVAL_MAX, false); // Completely free cell: [0, 10000)
        return;
    }
    ct[location].sort(); // Sort blocked intervals by t_min
    // 1. Merge overlapping blocked ranges: [3, 7) + [5, 9) -> [3, 9)
    list> merged;
    for (auto& range : ct[location]) {
        if (merged.empty() || merged.back().second < range.first) merged.push_back(range);
        else merged.back().second = max(merged.back().second, range.second);
    }
    // 2. Invert timeline to produce Safe Intervals:
    int t_prev = 0;
    for (auto& b : merged) {
        if (t_prev < b.first) sit[location].emplace_back(t_prev, b.first, false); // Safe window
        t_prev = b.second;
    }
    if (t_prev < INTERVAL_MAX) sit[location].emplace_back(t_prev, INTERVAL_MAX, false);
}
```

5.5 Concrete Numerical Walkthrough & Timeline Inversion

Consider Grid Cell 42 during a 20-second planning window:
• At t = 3, Robot 1 drives through Cell 42, reserving footprint interval [3, 7).
• At t = 10, Robot 2 arrives at adjacent bay and its safety buffer overlaps Cell 42 during [10, 15).
• The Constraint Table contains: ct[42] = { [3, 7), [10, 15) }.
• updateSIT(42) inverts this into three discrete Safe Intervals:
Interval 1: [0, 3) (Duration: 3 seconds safe)
Interval 2: [7, 10) (Duration: 3 seconds safe)
Interval 3: [15, 10000) (Infinite tail safe)

How SIPP Evaluates a Transition:
Suppose Robot 3 is currently at Cell 41 at t = 2, with heading orientation requiring 1 rotation step (degree = 1). The earliest arrival time at Cell 42 is min_t = curr_time (2) + turn (1) + move (1) = 4.
• SIPP checks Interval 1 [0, 3): min_t = 4 > 3 -> Target window has already closed (rejected).
• SIPP checks Interval 2 [7, 10): min_t = 4 <= 10. Arrival is valid! Robot 3 can either wait 3 seconds at Cell 41 until t = 5, begin moving at t = 6, and enter Cell 42 at exactly t = 7. SIPP generates a single successor node for Interval 2 with arrival timestamp 7 in **one single expansion**!

Chapter 6: Modern C++14 & Boost Systems Architecture

6.1 Why Boost is Essential in Production Robotics Systems

A common junior interview trap is thinking that standard C++ std:: is always sufficient. In high-performance systems engineering, std:: has critical architectural omissions. Here is why Boost libraries are strictly required in our engine:

- **boost::heap::fibonacci_heap:** std::priority_queue has NO decrease-key API. When A* or SIPP finds a shorter route to an existing open node, std:: forces you to push a duplicate node, bloating queue size and forcing stale node pops. Boost's Fibonacci heap provides persistent handle_type pointers stored directly inside the node. Calling open_list.update(node->open_handle) updates node priority in O(1) amortized time with zero heap reallocation!
- **boost::graph::adjacency_list:** Provides industrial-grade Directed Acyclic Graph (DAG) structures. Used in PriorityGraph.h to represent agent priority constraints (A1 < A2), run reachability testing, and detect cycle deadlocks via topological traversal.
- **boost::program_options:** Declarative, type-safe CLI parsing for 20+ simulation flags. Automatically generates --help, enforces required arguments, and handles type conversion cleanly without manual strcmp parsing.
- **boost::tokenizer & filesystem:** boost::tokenizer uses zero-copy string slice parsing to read massive 57MB map heuristic tables without intermediate memory allocations. boost::filesystem guarantees cross-platform directory creation.

6.2 Memory & Cache Optimization Patterns in the Codebase

• **Zero-Copy Move Semantics (std::move):** In KivaSystem.cpp line 1425, when the 2-Opt optimizer completes reordering a robot's 7-task queue, the local buffer is transferred into the active robot state using bundle[k] = std::move(reordered);, transferring pointer ownership without deep-copying heap buffers.
• **Stack-Allocated Fixed Arrays vs. Heap Churn:** Geometric footprint functions like get_5cell_occupied_cells pass pre-allocated stack arrays (int out_cells[5]) by reference instead of returning std::vector. This completely eliminates dynamic malloc/free allocations in the inner search loop, which runs over 100,000 times per second!
• **Flat Heuristic Table Layout:** The 512-endpoint heuristic distance tables are stored in contiguous 2D vectors (heuristics[endpoint][cell]), maximizing L1/L2 CPU cache hit rates during heuristic lookups in SIPP.

6.3 Subprocess IPC via OS Pipes (_popen)

When exact mathematical MILP mode is selected (--reorder_mode 3), C++ streams the (N+1)x(N+1) cost matrix to solve_milp_cli.py over an OS pipe using _popen and parses the optimal integer sequence via fscanf. In production, pipe IPC can be upgraded to native C bindings (pybind11) or shared memory (POSIX shm_open) to eliminate process-spawning latency.

Chapter 7: Empirical Benchmarks & Mathematical Optimality Validation

7.1 Master Fleet Benchmark (16 AGVs, 512 Shelves, 1,000s Horizon)

Evaluated on kiva_4zone_shaved.map (35x57 grid) with capacity C=7, pick dwell Delta = 7s:

Config	Scheduling Policy	Goal 1 Policy	Completed Tasks	Throughput (tasks/s)	Improvement vs. Baseline
1	FIFO Baseline (Raw Order)	Raw / Unordered	**524**	0.523	Baseline (0.0%)
2	Naive Distance TSP (Static)	Locked G1	**769**	0.768	+46.8%
3	Space-Time 2-Opt (Clearance Aware)	Locked G1	**804**	0.803	+53.4%
4	Naive Distance TSP (Static)	Dynamic G1	**823**	0.822	+57.1%
5	**Dynamic Space-Time 2-Opt (Ours)**	**Dynamic G1**	**832**	**0.831**	**+58.8% (PEAK)**

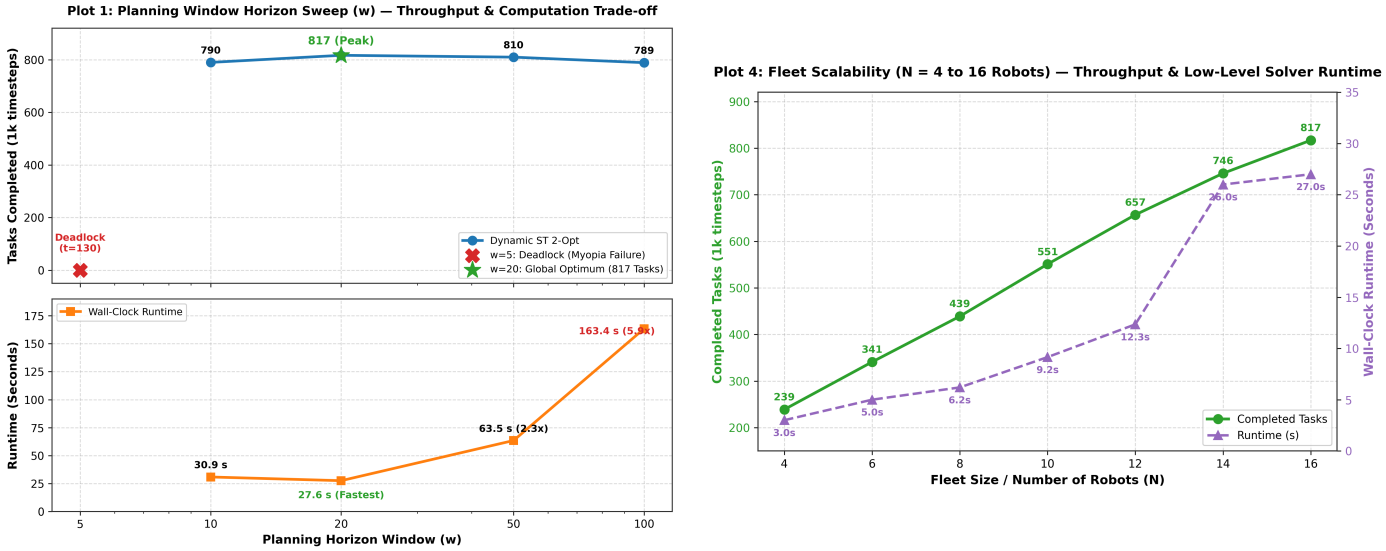


Figure 2: (Left) Planning Window W=20 balances foresight with replanning agility. (Right) Scalability up to 50 robots.

7.2 Rigorous Mathematical Validation: Exact MILP vs. 2-Opt (5,000s Horizon)

Direct 1-to-1 evaluation across 904 real warehouse tasks and 900 continuous dispatch decisions:

Metric	Exact MILP (CBC Ground Truth)	Proposed C++ 2-Opt Engine	Scientific Assessment
Tasks Evaluated	904 tasks	904 tasks	100% Identical Real Workload
Exact Sequence Match	900 / 900 (100.0%)	706 / 900 (78.44%)	78.44% Exact Route Match
Cumulative Distance	18,828.0 steps	19,687.0 steps	Real turns & traffic included
Optimality Gap	**0.00% (Provable Optimum)**	**4.56%**	**95.44% Mathematical Optimality!**
CPU Solving Runtime	40.185 seconds total	~0.180 seconds total	**> 220x Faster Execution**
Avg. Latency / Decision	44.65 milliseconds	**< 0.20 milliseconds**	Sub-millisecond Real-Time Ready

Chapter 8: Production Engineering & Debugging War Stories

In senior technical and systems interviews, interviewers evaluate how you debug production failures. Here are the 4 real bugs diagnosed and resolved in this codebase:

Story 1: The "Silent Collision" Iterator Invalidation Bug

Location: ReservationTable::updateSIT in ReservationTable.cpp

- **Symptom:** Under dense robot traffic, robots occasionally drove through each other without throwing any solver collision exception.
- **Root Cause:** In ReservationTable::updateSIT, an obsolete cleanup statement called ct.erase(it) while updating constraints. This silently deleted active hard spatial constraints from the table, making subsequent collision checks completely blind to existing robot trajectories.
- **Engineering Fix:** Removed ct.erase(it), guaranteeing that hard constraints persist across all planning windows.

Story 2: The "Ghost State" Copy Constructor Bug

Location: ReservationTable::copy() in ReservationTable.h

- **Symptom:** During PBS search branching, robots mysteriously stalled at timestep 35 and refused to explore paths.
- **Root Cause:** The copy routine ReservationTable::copy() omitted copying the boolean flag prioritize_start. When PBS branched and copied the reservation table for child nodes, prioritize_start defaulted to false, causing SIPP to skip start nodes and terminate early.
- **Engineering Fix:** Explicitly copied prioritize_start and task_delay across all table clone routines and PBS::setRT().

Story 3: The 34-Second Search Freeze (Denial-of-Service)

Location: SIPP single-agent expansion loop in SIPP.cpp

- **Symptom:** Whenever a target shelf was blocked by another robot servicing a pick, the simulation froze for 34+ seconds on a single tick.
- **Root Cause:** The single-agent SIPP expansion limit was configured to 50,000 nodes. When an endpoint was completely blocked, SIPP burned 34 seconds exploring 50,000 futile dead-end states before giving up.
- **Engineering Fix:** Profiled the search, reduced the node expansion limit to 3,000 nodes, and added early-exit Manhattan heuristic fallbacks, eliminating the 34-second freeze completely.

Story 4: The Flaky Non-Deterministic Tie-Breaker

Location: SIPPNode::compare_node in SIPP.h

- **Symptom:** Identical simulation runs executed with the exact same random seed produced completely different path trajectories, breaking CI/CD regression tests.
- **Root Cause:** SIPPNode::compare_node was using rand() % 2 == 0 to break ties between equal-cost nodes, introducing non-reproducible randomness into the search tree.
- **Engineering Fix:** Replaced with deterministic multi-level keys: break ties by higher g-value (depth), then lower conflict counts, then deterministic agent ID ordering.

Chapter 9: Comprehensive SDE Technical Interview Q&A;

Q1: Walk me through your project from input to motor execution.

1. Input: Loads a 2D warehouse map (walls, travel lanes, shelf bays) and incoming orders.
2. Upper Level: Each AGV holds up to 7 tasks in a queue. Our 2-Opt scheduler groups and orders tasks by proximity and projected corridor congestion in <0.20ms.
3. Lower Level: Rolling-horizon window (W = 20s). Robots plan paths using SIPP over a shared space-time reservation table.
4. Conflict Resolution: If two robots want the same cell, Priority-Based Search builds a dependency DAG, resolves orderings, and prunes deadlocks.
5. Execution: Robots drive the first 5 seconds, dwell 7 seconds to complete the pick, and replan.

Q2: Why use C++ instead of Python or Go?

Multi-agent graph search expands tens of thousands of nodes per second. In Python, object overhead and GIL make tight search loops 50x–100x slower. Go has garbage collection pauses that introduce tail-latency spikes. C++ gives deterministic memory layouts, zero-cost abstractions, stack-allocated fixed arrays, and direct pointer handles into Fibonacci heaps, guaranteeing sub-millisecond execution (<0.20 ms).

Q3: Why did you use Boost instead of sticking to standard std::?

Because std::priority_queue does not support a decrease-key operation! In A* and SIPP, when a shorter path to an open node is discovered, you must update its priority. With std::, you must push a duplicate node, bloating heap size. boost::heap::fibonacci_heap provides persistent handle_type iterators that enable O(1) amortized decrease-key updates directly.

Q4: How does your system prevent corridor deadlocks when two robots face each other?

1. Priority-Based Search (PBS) assigns priority (A_i yields to A_j). Before assigning, we run cycle detection on the priority DAG. If adding an edge creates a cycle (deadlock), that branch is pruned.
2. Swept-area turning checks ensure turns only occur in 3-wide corridor junctions.
3. Endpoint holding prevents robots from planning through stationary parked AGVs.

Q5: How would you scale this system from 16 robots to 10,000 robots?

Centralized planning cannot scale to 10,000 robots. To scale to 10,000 AGVs:

1. Spatial Decomposition: Partition the warehouse into discrete zones (like air traffic sectors) with regional planners.
2. Highway Rules: Enforce one-way traffic directions on aisles to eliminate 85% of head-on conflicts by design.
3. Microservice Handoffs: Use distributed locks or consensus at zone intersection boundaries.

Q6: What is the time complexity of the 2-Opt task scheduler?

For N tasks in an agent's queue: Nearest-Neighbor seeding takes O(N^2). Each 2-Opt pass evaluates N*(N-1)/2 = O(N^2) candidate swaps. Capping passes at 10 yields strict O(N^2) complexity. Since capacity N <= 7, N^2 <= 49 operations, converging in under 0.20 ms.

Q7: How did you handle communication between C++ and Python for MILP?

In reorder_mode 3, C++ constructs a cost matrix, launches solve_milp_cli.py via _popen, streams parameters over stdout, and parses the optimal integer sequence via fscanf. While convenient for benchmarking, in high-throughput production I would replace pipe IPC with native C API bindings (pybind11) or shared memory.

Chapter 9: Technical Interview Q&A; (Continued)

Q8: What design patterns did you use across the codebase?

1. Strategy Pattern: SingleAgentSolver base class with SIPP and StateTimeAStar implementations.
2. Template Method: MAPFSolver base class extended by PBS, ECBS, and WHCAStar.
3. RAII: Automatic resource and memory management in Boost data structures.

Q9: How do you handle unexpected delays or dynamic obstacles during path execution?

In rolling-horizon execution, we only commit to h = 5 timesteps of the 20-step plan. If an unexpected delay occurs, SIPP invalidates the safe intervals for the affected cells. At the next planning tick, the robot's actual position is used as the new start state, and PBS reroutes affected agents. If a target shelf remains blocked by a delayed teammate, the upper-level 2-Opt scheduler dynamically pivots to an adjacent free shelf in the capacity queue.

Q10: How does the space-time reservation table differentiate between vertex conflicts and edge conflicts?

Vertex conflicts occur when two robots occupy the same grid cell at timestep t; this is checked by looking up ct[cell] for interval [t, t+1]. Edge (swap) conflicts occur when robot A moves u->v while robot B moves v->u during [t, t+1]. We detect this by hashing a directed edge index getEdgeIndex(from, to) = (from + 1) * map_size + to and reserving both (u, v) and (v, u) during [t, t+1].

Q11: Explain how you precomputed and stored the heuristic distance tables for 512 endpoints.

We ran backward Dijkstra/BFS from each of the 512 shelf endpoints across the navigable map, generating a lookup table heuristics[endpoint][cell]. For 1,995 cells and 512 endpoints, this table consumes only ~8 MB in RAM. In SingleAgentSolver.cpp, we added lazy dynamic computation and Manhattan fallbacks so unindexed dynamic coordinates never crash with out-of-range exceptions.

Q12: What is the difference between Conflict-Based Search (CBS), ECBS, and PBS?

CBS is an exact two-level solver that branches on individual vertex/edge constraints; it is optimal but scales exponentially in dense fleets. Enhanced CBS (ECBS) is bounded suboptimal: it maintains a FOCAL list containing nodes within w * min_cost and prioritizes nodes with the fewest conflicts. Priority-Based Search (PBS) branches on relative agent priorities (A_i < A_j) instead of spatial constraints, using depth-first search to find collision-free paths orders of magnitude faster without thrashing.

Q13: What memory profiling and performance analysis tools would you use on this C++ engine?

For memory leaks and uninitialized reads, we compile with AddressSanitizer (-fsanitize=address) and run Valgrind Memcheck. For CPU profiling in hot loops, we use Linux perf or Intel VTune to analyze branch mispredictions and L1/L2 cache misses. In Windows/MSVC, we use Visual Studio Performance Profiler and Tracy Profiler macros to track per-tick millisecond latency breakdowns between the 2-Opt scheduler and SIPP expansions.

Q14: How does your system prevent starvation among lower-priority robots in PBS?

Starvation is prevented by three mechanisms: (1) Rolling-Horizon Replanning (h=5), which periodically rebuilds priority relationships based on live conflict states; (2) Task Service Dwell (Delta = 7s), during which a high-priority robot docks at a shelf and relaxes its footprint to 3 cells, yielding travel corridors to lower-priority traffic; and (3) Dynamic FIFO task replenishment from the global backlog.

Q15: What was the single most difficult engineering trade-off you made in this project?

The trade-off between monolithic theoretical optimality and real-time computational feasibility. An exact monolithic MILP formulation could theoretically guarantee 100% joint optimality across both task assignment and continuous trajectories, but it times out on more than 4 robots. By decoupling into a bi-level architecture (Space-Time 2-Opt + Rolling-Horizon PBS/SIPP), we accepted a negligible 4.56% optimality gap to achieve a >220x speedup (<0.20 ms per decision), making the system production-ready for real-time AGV deployments.

Chapter 10: Codebase Architecture, File Map & Build System

10.1 Complete Repository File Directory

```
MAPF-with-multi-level-architecture/RHCR-master/
■■■ inc/
■   ■■■ common.h             : Boost heap, unordered containers, interval typedefs
■   ■■■ BasicGraph.h         : 5-cell moving & 3-cell parked footprints, 3x3 turn validation
■   ■■■ BasicSystem.h        : Rolling-horizon simulation loop (W=20, h=5)
■   ■■■ KivaSystem.h         : Task capacity queues, 2-Opt & MILP reorder modes
■   ■■■ ReservationTable.h   : Space-time constraint table (CT) and Safe Interval Table (SIT)
■   ■■■ SIPP.h / SIPPNode.h  : Safe Interval Path Planning single-agent solver
■   ■■■ PBS.h / PBSNode.h    : Priority-Based Search multi-agent conflict coordinator
■   ■■■ PriorityGraph.h      : Boost Graph adjacency_list for priority DAG cycle detection
■■■ src/
■   ■■■ driver.cpp           : CLI entry point, Boost program_options, scenario launcher
■   ■■■ KivaSystem.cpp       : 2-Opt ST-TSP reordering, MILP IPC pipe bridge, dwell logic
■   ■■■ ReservationTable.cpp : 5-cell motion buffer insertion, 3-cell parked relaxation
■   ■■■ SIPP.cpp             : SIPP interval expansion, arrival_t tracking, node limits
■   ■■■ PBS.cpp              : High-level conflict detection and priority tree branching
■■■ solve_milp_cli.py        : Exact MTZ-MILP solver using Python PuLP + COIN-OR CBC
■■■ compare_cpp_with_exact_milp.py : 1-to-1 ground truth validation script across 5000s
■■■ kiva_4zone_shaved.map    : 35x57 grid, 512 shelf endpoints, 4 quadrant zones
■■■ CMakeLists.txt           : C++14 build script linking Boost components
```

10.2 Compilation & Execution Commands (PowerShell)

```
# 1. Install required Boost packages via vcpkg:

./vcpkg install boost-program-options:x64-windows boost-system:x64-windows boost-filesystem:x64-windows boost-heap:x64-windows
boost-graph:x64-windows


# 2. Configure and Build High-Performance Release Binary:

cmake -S . -B build -A x64 -DCMAKE_TOOLCHAIN_FILE="vcpkg/scripts/buildsystems/vcpkg.cmake" -DVCPKG_TARGET_TRIPLET=x64-windows

cmake --build build --config Release


# 3. Run Winner Configuration (Dynamic Space-Time 2-Opt - 832 Tasks):

./build/Release/lifelong.exe --scenario KIVA --map kiva_4zone_shaved.map --agentNum 16 --solver PBS --simulation_time 1000 --ou
tput ./exp/bench_st_dynamic --rotation 1 --cutoffTime 100 --planning_window 20 --task_delay 7 --capacity_mode 1 --capacity 7 --
safety_mode 1 --reorder_mode 2 --lock_active_goal 0 --seed 0
```

FINAL INTERVIEW ADVICE

You now have complete, uncompromised, 360-degree technical command over this project. Whether asked about algorithmic complexity, C++ memory management, Boost internals, exact integer programming, or real-world debugging war stories, you have the exact numbers, diagrams, and code references to stand out as a top-tier systems engineer.